

1.write a program for harshad number

```
num=18
save=num
sum=0
while num!=0:
    r=num%10
    sum=sum+r
    num=num//10

if save%sum==0:
    print(save," is a harshad number")
else:
    print("not harshad nuber")
```

18 is a harshad number

2.write a python program to find the union of two sets

```
n1=int(input("enter the size n1  "))
a=[int(input("enter the values  ")) for i in range(n1)]
n2=int(input("enter the size n2  "))
b=[int(input("enter the values  ")) for i in range(n2)]

res=[]

for i in a:
    res.append(i)
for i in b:
    if i not in res:
        res.append(i)
print(res)
```

```
enter the size n1  5
enter the values  1
enter the values  2
enter the values  3
enter the values  4
enter the values  5
enter the size n2  5
enter the values  6
enter the values  7
enter the values  1
enter the values  2
enter the values  3
[1, 2, 3, 4, 5, 6, 7]
```

3.write a python program to print the common elements in the list

```
n1=int(input("enter the size n1  "))
a=[int(input("enter the values  ")) for i in range(n1)]
n2=int(input("enter the size n2  "))
b=[int(input("enter the values  ")) for i in range(n2)]

res=[]

for i in a:
    if i in b:
        res.append(i)
print("comman elemantsare ",res)
```

```
enter the size n1  4
enter the values  7
enter the values  17
enter the values  18
enter the values  47
enter the size n2  4
enter the values  3
enter the values  7
enter the values  17
enter the values  18
comman elemantsare  [7, 17, 18]
```

4.write a python program to print all the unique elements present in a but not in b

```
n1=int(input("enter the size n1  "))
a=[int(input("enter the values  ")) for i in range(n1)]
n2=int(input("enter the size n2  "))
```

```
n2=int(input("enter the size n2 "))  
b=[int(input("enter the values ")) for i in range(n2)]  
  
res=[]  
  
for i in a:  
    if i not in b:  
        res.append(i)  
print("unique elemnts are ",res)
```

```
enter the size n1 4  
enter the values 1  
enter the values 2  
enter the values 3  
enter the values 4  
enter the size n2 4  
enter the values 6  
enter the values 7  
enter the values 1  
enter the values 2  
unique elemnts are [3, 4]
```